

ADVANCED TOPICS OF MACHINE LEARNING

Course Description

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Universidad Distrital Francisco José de Caldas

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 You don't know who I am
- 2 Course Overview
- 3 Grading & Rules
- 4 Bibliography



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Academic Experience

- **Computer Engineer**, M.Sc. in Computer Engineering, and *researcher* for **17 years**.
- 9 years as **full-time associate professor** at colleges, in **Computer Engineering programs**.
- 3 years as **lecturer professor** for both colleges and **government STEM programs**.
- Speaker at **IEEE** events and colleges in Colombia, Brazil, and Bolivia.



Non-academic Experience



- **PyCon Colombia** and **Python Bogotá** co-organizer.
- 3 years as **software engineer** for several **tech companies** in Colombia.
- 3 years as **Technical Leader** of **Machine Learning and Data Science** at a USA startup.
- 1.5 years as **MLOps Engineer** for a **LATAM Fintech** company.
- Currently, **Senior Engineering Manager** of **Data Engineering** and **Machine Learning** at Blend 360.



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Overview

This course is designed to **introduce undergraduate students** to **advanced machine learning techniques** as part of the foundation for becoming an experienced *AI engineer* capable of developing *intelligent solutions*.

The course starts with a **comprehensive study** of **reinforcement learning** and **generative models**. Then, it transitions into **advanced unsupervised learning** and **deep learning optimization techniques**. Finally, we will focus on **online learning**, **transfer learning**, and **model interpretability** for both **research** and **industry applications**.

Classes will consist of **theoretical lectures**, **algorithm implementation sessions**, and **cutting-edge research projects**. Also, you must complete some readings from *advanced ML research papers* and *deep learning*. In addition, there will be a **semester-long research project**, **one final course test**, and **four practical challenges**.



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mini-projects

2



Goals

The **main goal** of this course is to **provide students** with **advanced concepts** and **cutting-edge techniques** for **intelligent systems design** capable of analyzing data, optimizing processes, and improving **human-machine interactions**.

At the end of this course you should be able to **implement advanced ML models** including reinforcement learning, generative models, and **transfer learning techniques**. Also, you should be able to **develop** complete **AI solutions** that solve real-world problems using **state-of-the-art algorithms** and optimization techniques.



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LLMs → Foundational
Tune + RAG



Prerequisites

This is an advanced course, so you must have some knowledge in:

- **Programming** in **Python** (advanced level), including libraries like **NumPy**, **Pandas**, and **Scikit-learn**.
- **Mathematics** fundamentals: **linear algebra**, **calculus**, and **probability theory**.
- **Statistics** and **statistical inference**: hypothesis testing, distributions, and regression analysis.
- **Basic Machine Learning** concepts: supervised/unsupervised learning, model evaluation, and cross-validation.
- **Git** usage, **Jupyter Notebooks**, and experience with ML frameworks like **TensorFlow** or **PyTorch**.
- **Recommended**: Experience with Docker containers and research paper reading.



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Syllabus

Period	Topic	Time
Period I	Introduction to Machine Learning	3 sessions
	Mathematics for Machine Learning	3 sessions
	Neural Networks & Deep Learning Fundamentals	6 sessions
	Reinforcement Learning Fundamentals	9 sessions
	Generative Models	5 sessions
	Mid-Term Test	1 session
Period II	Probabilistic Models	3 sessions
	Advanced Unsupervised Learning	6 sessions
	Online Learning + Transfer	3 sessions
	ML Interpretability Techniques	3 sessions
	Final Test	1 session
Period III	Project Dissertation	3 sessions



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Grades Percentages

Period	Item	Percentage
Period I	Mini-Projects	15%
	Middle-Term Test	20%
Period II	Mini-Projects	15%
	Final Test	20%
Period III	Paper + Poster	15%
	Report + Implementation	10%
	Presentation	5%

30%
40%
30%



Don't hate the player, hate the game

- All assignments must be submitted **handwritten**, **on time**, and in **English**. Grammar and spelling will **not** be evaluated.
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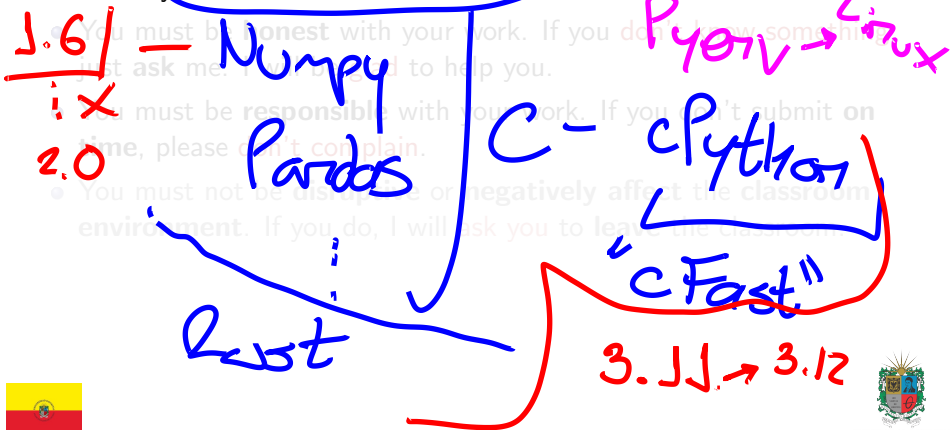
Code of Conduct

- **Always** be **respectful** to your **classmates** and to me. You must be **kind** to everyone inside (*and outside*) the classroom.
- There is **no** best **programming language**, **tool**, or **technology**. There are only **better** or **worse** solutions.
- You must be **honest** with your work. If you **don't know something**, just **ask** me. I will be **glad** to help you.
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Moodle



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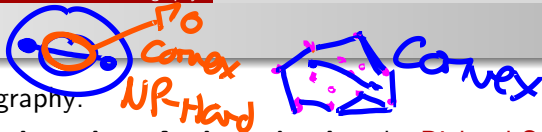


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Bibliography



Recommended bibliography.

- **Reinforcement Learning: An Introduction**, by Richard S. Sutton & Andrew G. Barto.
- **Deep Learning**, by Ian Goodfellow, Yoshua Bengio, & Aaron Courville.
- **Generative Deep Learning**, by David Foster.
- **Pattern Recognition and Machine Learning**, by Christopher M. Bishop.
- **The Elements of Statistical Learning**, by Trevor Hastie, Robert Tibshirani, & Jerome Friedman.
- **Interpretable Machine Learning**, by Christoph Molnar.
- **Online Learning and Online Convex Optimization**, by Shai Shalev-Shwartz.
- **Bayesian Reasoning and Machine Learning**, by David Barber.



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Thanks!

Questions?



My Profile: www.linkedin.com/in/casierrav

